NLP Foundations

Spread

- NLP in GenAI context
- Text representation
- Language modelling basics
- Classical NLP pipeline
- Bridge to GenAI/Agents

NLP in GenAI Context

- Interface layer of AI systems
- Chatbots, Search, Automation
- Real-world: Support, Finance, Healthcare

Text Representation

- Bag of Words → TF-IDF → Word2Vec → Embeddings
- Goal: Convert text → numbers

Text Representation Intuition

- Similar meaning → similar vectors
- Example: king – man + woman ≈ queen

Language Modeling Basics

- Predict next word
- Context window
- Probability distribution

Language Models Evolution

- N-grams → RNN → Transformers
- Parallel processing + context awareness

Classical NLP Pipeline

- Cleaning → Tokenization → Stopwords → Lemmatization → Features → Model

Hands-on Pipeline

- Tokenize text using NLTK
- Apply TF-IDF
- Build simple classifier

Limitations of Classical NLP

- No deep context
- Manual features
- Poor generalization

Bridge to GenAI

- Classical NLP → Deep Learning → LLMs → Agents
- Agent Loop: Observe → Reason → Act

Modern NLP Architecture

- Input → Embeddings → Retrieval → LLM → Output

Hands-on Task

- Build mini NLP pipeline
- Compare TF-IDF vs Embeddings

Key Takeaways

- NLP powers GenAI
- Representation matters
- LLMs are prediction engines